tRNAstudio: facilitating the study of human tRNA-seq datasets

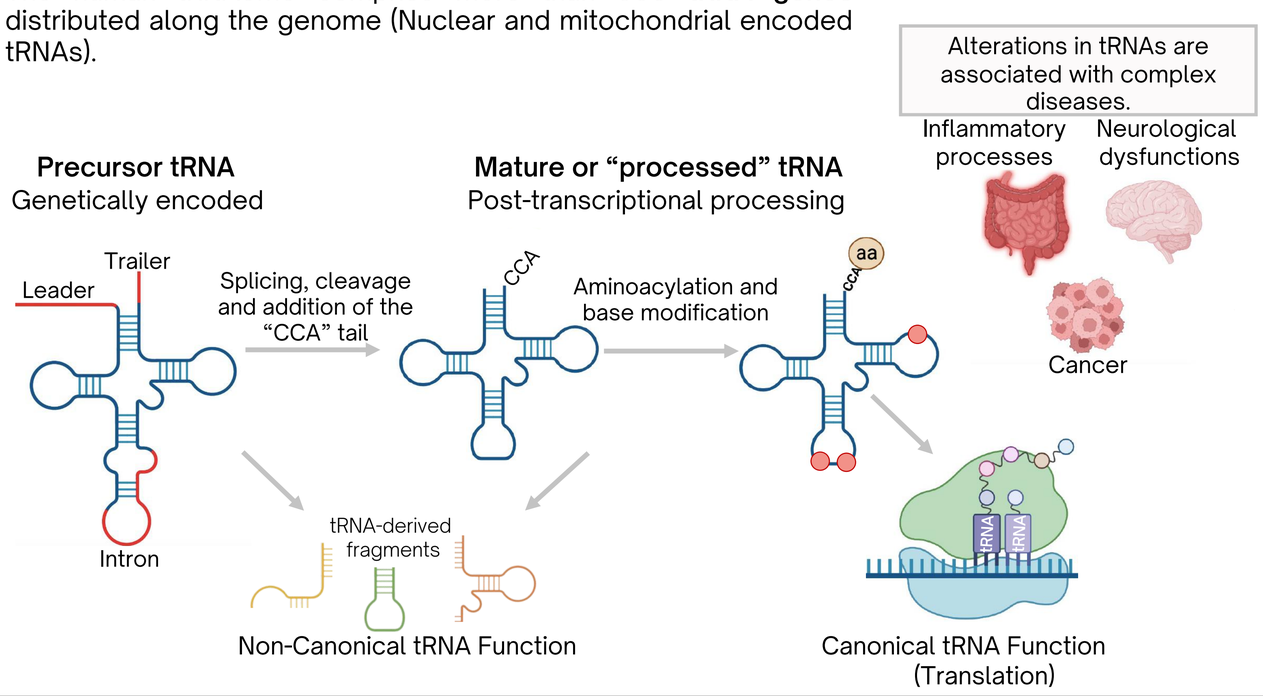
Marina Murillo-Recio [1], Adrian Gabriel Torres [1], Lluís Ribas de Pouplana [1,2]
[1] Institute for Research in Biomedicine, The Barcelona Institute of Science and Technology, Barcelona, Catalonia, 08028, Spain.
[2] Catalan Institution for Research and Advanced Studies, Barcelona, Catalonia, 08010, Spain.

ABSTRACT

High-throughput sequencing of transfer RNAs (tRNA-Seq) is a powerful approach to characterize the cellular tRNA pool. Currently, however, analysing tRNA-Seq datasets requires strong bioinformatics and programming skills. **tRNAstudio facilitates the analysis of tRNA-Seq datasets and extracts information on tRNA gene expression, post-transcriptional tRNA modification levels, and tRNA processing steps.** Users need only running a few simple bash commands to activate a graphical user interface that allows the easy processing of tRNA-Seq datasets in local mode. Output files include extensive graphical representations and associated numerical tables, and an interactive html summary report to help interpret the data. We have validated tRNAstudio using datasets generated by different experimental methods and derived from human cell lines and tissues that present distinct patterns of tRNA expression, modification and processing.

THE COMPLEXITY OF tRNA BIOLOGY

The human tRNAome comprise more than **600 tRNA genes** distributed along the genome (Nuclear and mitochondrial encoded tRNAs).

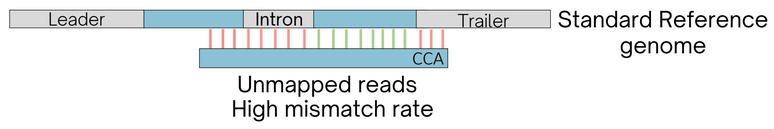


Transfer RNAs (tRNAs) are non-coding RNAs that bring amino acids to the ribosome and are essential for protein synthesis. tRNAs are the most highly chemically modified RNA in the cell. tRNAs can be processed into tRNA-derived fragments (tRFs) that perform a wide range of roles beyond protein synthesis.

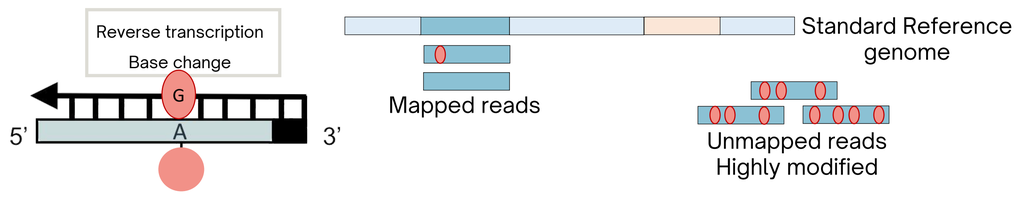
tRNA-seq IS THE GOLD STANDARD METHOD TO STUDY tRNAs BUT...

- Reads derived from processed tRNAs do not match the reference genome.

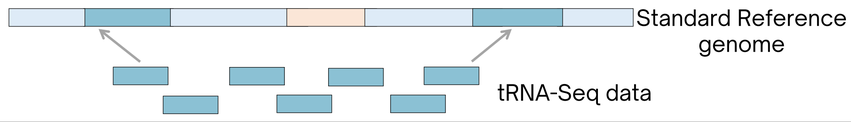
Reads with **addition/deletion of tRNA nucleotides** prevent read alignment.



Reads that come from highly **modified tRNA introduce discrepancies** (mismatches).



- Sequence similarity between different tRNAs produces **multimapping**.



It is possible to study tRNA biology by means of high throughput sequencing (tRNA-Seq). The presence of tRNA modifications, the sequence similarity between different tRNAs, and the large number of tRNAs encoded in the genome impair the interpretation and downstream analysis of tRNA-Seq datasets.

tRNAstudio A FULLY AUTOMATED PIPELINE DESIGNED TO ANALYZE tRNA-Seq

1 MAPPING STRATEGY: tRNAstudio performs different rounds of alignment using different reference genomes.

Generation of artificial tRNAs genomes improve tRNA detection and classification.

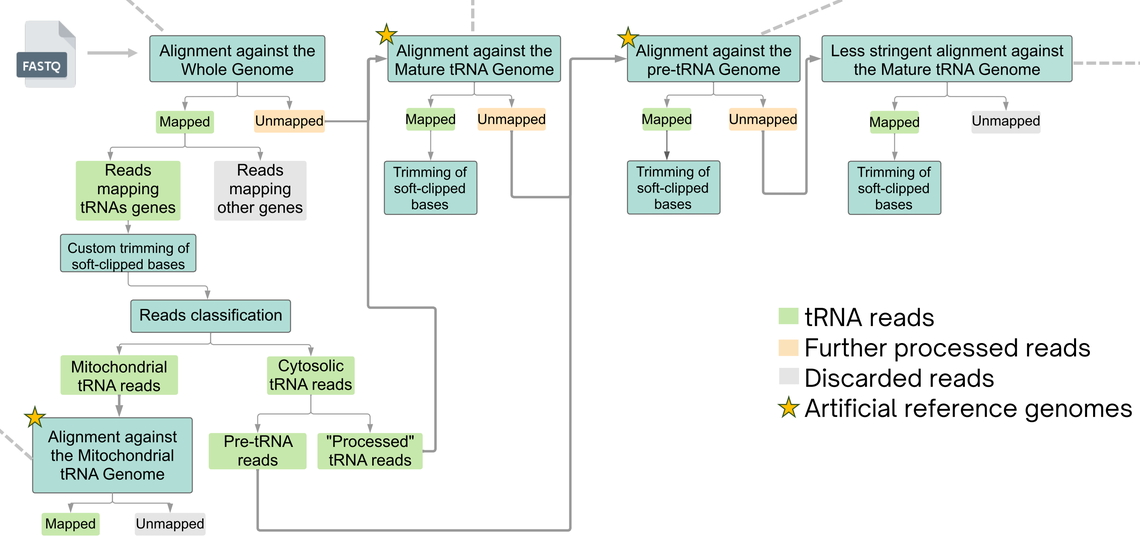
Crucial step to filter reads that do not map defined tRNAs sequences.

Artificial "genome" with only nuclear encoded tRNA sequences without 5′-leader, 3′-trailer and intronic regions. Addition of a "CCA" at the 3'-end. Sequences with 100% of identity are grouped into families in order to **control multimapping.**

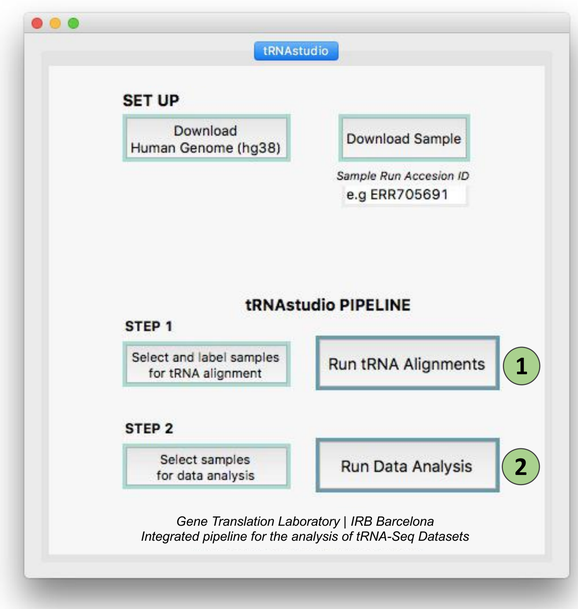
Artificial "genome" with only nuclear encoded tRNA sequences containing 5′-leader, 3′-trailer and intronic regions.

Reducing alignment stringency to allow the **detection of highly modified tRNAs.**

Artificial "genome" with only mitochondrial encoded tRNA sequences.

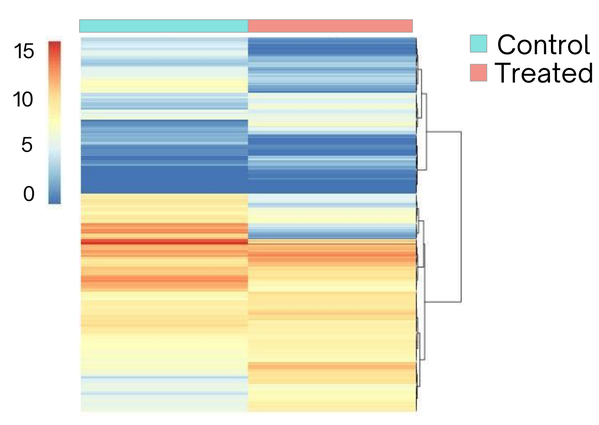


PIPELINE INTEGRATION
Packaged into a user-friendly graphical user interface (GUI).

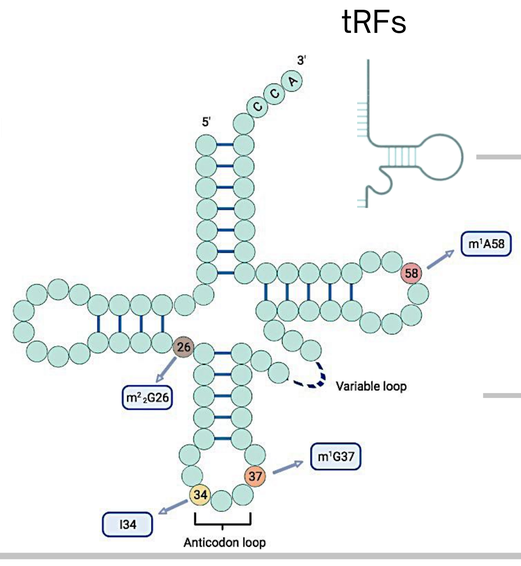


2 DATA ANALYSIS REPORT
Interactive html summary report and extensive graphical representations of the data.

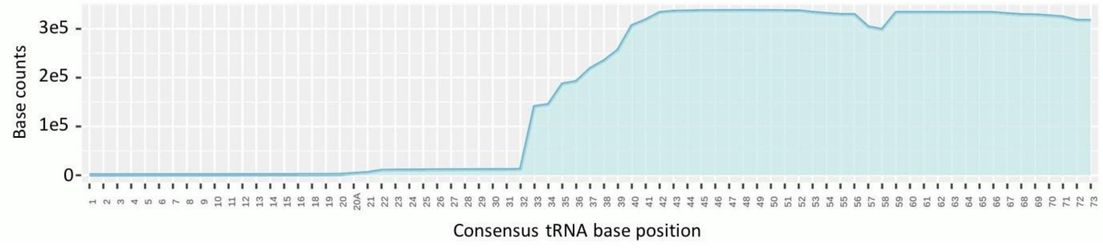
Differential expression analysis of tRNAs



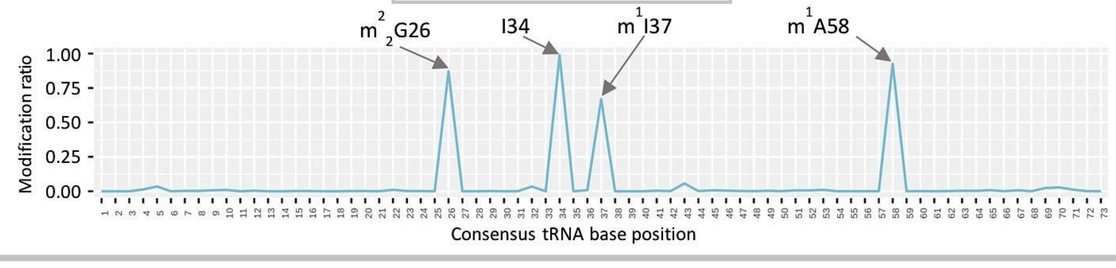
tRFs



tRNA gene coverage



tRNA modifications



INSTITUTE FOR RESEARCH IN BIOMEDICINE
IRB BARCELONA

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GitHub tRNAstudio

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@MarinaBioinfo
marina.murillo@irbbarcelona.org